

FilterNet: A Filtered Gait Motion Fusion Network for Classifying Adult Spinal Deformity

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Abstract—Analyzing periodic human motion from videos is vital for applications such as action recognition and healthcare. In gait analysis, models must capture subtle phase-specific motion patterns. However, processing entire sequences often introduces noise and reduces accuracy. We propose FilterNet, a frame-selection framework that identifies phase-relevant frames to improve temporal modeling and enhance discriminative representation. On a clinical gait video dataset, FilterNet achieves 74.5% accuracy, 75.2% precision, and 74.3% F1-score, outperforming baseline methods. Though demonstrated in a medical context, the framework is broadly applicable to other periodic motion analysis tasks. Code is available at: https://github.com/ChenKaiXuSan/FilterNet_ASD_PyTorch.

Index Terms—Medical video analysis, Gait classification, Key frame selection, Phase-aware learning, Clinical pose selection

I. INTRODUCTION

Analyzing periodic motion in videos is essential for applications such as action recognition, sports analytics, and healthcare. Gait, as a typical periodic motion, exhibits subtle phase-dependent dynamics that require fine-grained temporal modeling. Recent studies have proposed various approaches for gait representation. [1] demonstrated the clinical potential of deep gait-based health assessment. In medical contexts, [2], [3] introduced diagnostic frameworks leveraging phase-based fusion to capture gait symmetry and periodicity.

However, many methods process all frames uniformly, neglecting phase relevance. Such sampling introduces redundancy and motion artifacts, resulting in noisy representations and limited interpretability. A phase-aware strategy is therefore crucial to emphasize diagnostically informative motion cues and improve the reliability of temporal modeling.

To analyze the impact of frame selection strategies, we examine motion variation within the left swing phase, as shown in Figure 1. The top row displays four consecutive frames along with relevance scores predicted by **FilterNet**. The left swing phase corresponds to the period when the left foot is airborne and the right foot supports the body. To quantify motion diversity, we overlay the selected frames using pose keypoints and compute the arm-to-body angle θ as a proxy for postural variation. Figure 1(a) shows a uniform sampling strategy selecting Frame 0 and Frame 2, which have similar limb configurations and limited variability. In contrast, Figure 1(b) shows the FilterNet-based strategy selecting

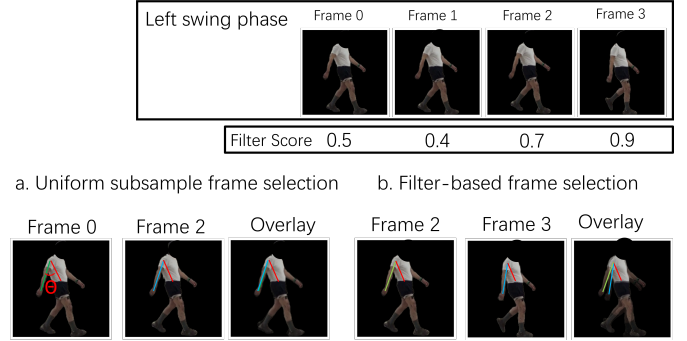


Fig. 1: Comparison of frame selection strategies during the left swing phase. (a) Uniform sampling selects evenly spaced frames. (b) Our method selects frames with higher diagnostic relevance based on FilterNet scores.

Frame 2 and Frame 3, capturing more distinct postural states and reflecting greater intra-phase motion diversity.

To support interpretable frame selection in periodic motion analysis, we propose a two-step framework centered on **FilterNet**. In **Step 1**, FilterNet is trained to assign a relevance score ($[0, 1]$) to each frame based on gait-specific cues. In **Step 2**, the top T frames per gait phase are selected and passed to a downstream classifier.

This decoupled design enables efficient, phase-aware modeling by focusing on diagnostically salient frames. Unlike uniform sampling or implicit attention, our method explicitly filters for informative motion patterns. While evaluated on clinical gait data, the approach is generalizable to other periodic video tasks.

The main contributions of this paper are as follows:

- We propose a two-step framework for periodic motion analysis that performs phase-aware frame selection and temporal segmentation, improving the representation of motion dynamics in video sequences.
- We introduce **FilterNet**, a lightweight frame-scoring model that selects informative frames based on visual cues within each motion phase, enhancing feature discriminability for downstream tasks.

II. RELATED WORK

A. Gait Analysis and Periodic Motion Modeling

Gait, as a canonical example of periodic human motion, has been extensively studied across domains such as action recognition, biometric identification, surveillance, and healthcare. It is characterized by cyclic and phase-dependent motion patterns, which require temporal modeling techniques that can capture subtle intra-cycle variations. Traditional gait analysis methods often rely on radiographic imaging [4] or 3D motion capture systems [5], which provide precise structural or kinematic data. However, these methods are limited by the need for controlled environments, expensive equipment, and time-consuming manual procedures. To address these limitations, recent research has shifted toward video-based approaches that use 2D pose estimation or direct appearance modeling to extract gait features in more accessible settings.

B. Video-based Gait Classification

With the rise of deep learning, direct modeling of gait from raw video frames has gained traction. These methods aim to learn spatiotemporal representations without requiring explicit pose extraction or motion capture data. [2] propose a two-stage CNN pipeline for diagnosing adult spinal deformity (ASD) from gait sequences, using temporal convolution and frame-level fusion. Similarly, [3] introduce PhaseMix, a temporal fusion method that emphasizes periodicity and gait symmetry across video frames. Although these methods achieve promising performance in medical gait classification, they generally apply uniform frame sampling or fixed-length segmentation. This approach treats each frame as equally informative, failing to exploit the semantic or diagnostic significance of specific gait phases. As a result, low-quality or redundant frames can degrade the discriminative power of temporal features and reduce model efficiency.

C. Frame Selection and Temporal Attention

Frame selection and temporal attention mechanisms have been widely explored in general video understanding tasks, including action recognition [6], event detection [7], and video summarization [8]. These methods aim to identify salient or task-relevant frames while discarding background or low-information content. Temporal attention networks dynamically assign weights to frames based on their contextual importance, whereas keyframe selection strategies explicitly rank and sample frames that maximize visual or semantic diversity. For example, Girdhar et al. [8] introduce a transformer-based model that localizes key moments in videos for action classification.

Inspired by these ideas, our work introduces a phase-aware frame filtering strategy specifically designed for periodic motion. Rather than treating frames uniformly, our method scores each frame based on its motion-phase context and visual relevance, selecting only those that best represent intra-phase variation. This enables more compact and discriminative temporal representations, particularly useful for downstream classification tasks in resource-constrained or clinical settings.

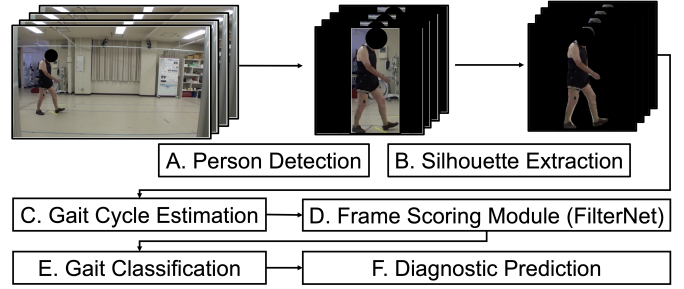


Fig. 2: The flow chart of the proposed gait-based diagnostic framework.

III. METHOD

A. Overview

The core novelty of our framework lies in its two-step design. In **Stage 1**, we train **FilterNet**, a lightweight scoring network that learns to assign frame-level diagnostic relevance based on phase-specific gait cues. In **Stage 2**, only the top-ranked frames are retained for downstream classification. This decoupled approach enables interpretable, phase-aware video modeling and improves robustness by filtering out noisy or redundant inputs.

Figure 2 illustrates the full pipeline of our gait-based diagnostic framework, which consists of six modular components (A–F). **(A) Person Detection:** A person detection module localizes the subject in each frame of the input video, suppressing background noise and ensuring spatial consistency. **(B) Silhouette Extraction:** The detected bounding boxes are refined using a silhouette segmentation model to extract clean human contours. This preserves structural cues such as limb configuration and body symmetry. **(C) Gait Cycle Estimation:** The silhouette sequence is temporally segmented into left and right swing phases based on gait cycle analysis [3], allowing for consistent temporal alignment across samples. **(D) Frame Scoring via FilterNet:** Each frame is evaluated by *FilterNet*, which predicts a continuous relevance score within each phase. The top-scoring frames are selected as diagnostically meaningful keyframes. Further details are provided in Sec. III-B. **(E) Gait Classification:** A spatiotemporal model aggregates features across selected keyframes to perform disease classification. Phase-aware frame selection improves efficiency and enhances clinical interpretability.

(F) Diagnostic Prediction: The model outputs a predicted disease label based on learned gait representations, supporting non-invasive clinical screening.

This coarse-to-fine framework not only improves classification accuracy but also enhances explainability by focusing on phase-specific salient frames. While validated on clinical gait analysis, the design is generalizable to other structured motion tasks such as sports performance assessment, rehabilitation monitoring, or skill evaluation in instructional videos.

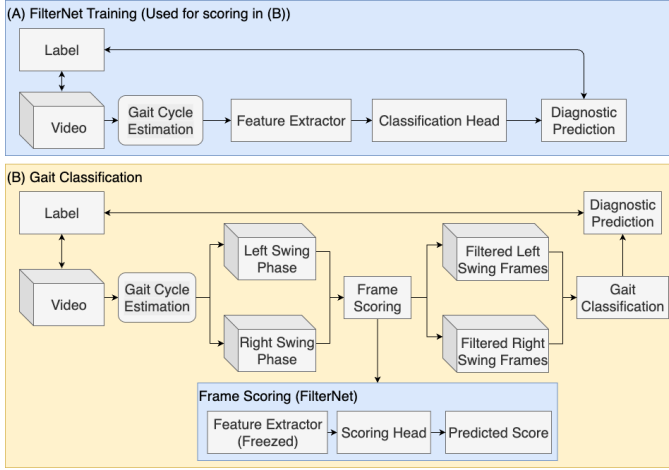


Fig. 3: Overview of the proposed framework. (A) FilterNet is trained to predict diagnostic labels from gait video frames. (B) During inference, each video is segmented into gait phases, and the pretrained FilterNet selects high-relevance frames for final disease classification.

B. FilterNet

To identify diagnostically relevant frames from gait videos, we propose **FilterNet**, a frame-wise scoring network that selects key frames for downstream classification. An overview of FilterNet is illustrated in Figure 3 (A). The model is based on a ResNet-101 backbone originally designed for image classification. To adapt it for regression, we replace the final fully connected layer with a single-node output layer that predicts a scalar score in the range $[0, 1]$, representing the diagnostic relevance of each frame.

During inference, each frame within a gait phase is processed independently through FilterNet to obtain its relevance score. The top- T scoring frames are then selected to represent each gait phase (i.e., left swing and right swing), allowing the model to emphasize high-saliency moments that are most indicative of abnormal gait patterns.

This filtering strategy effectively removes redundant frames and preserves diagnostically informative ones, thereby enhancing the performance of the subsequent gait classification module.

C. Gait Classification

The gait classification pipeline consists of two sequential steps: (1) FilterNet-based frame scoring, and (2) classification using the selected key frames. An overview of this process is shown in Figure 3 (B).

1) *Step 1: Score Prediction by FilterNet*: In the first step, we apply the FilterNet model pretrained as described in Section III-B, with its weights frozen during inference. The classification head is replaced with a scoring head to output a relevance score for each frame. Each frame within a gait phase is independently passed through FilterNet to obtain its diagnostic relevance score. To maintain temporal coherence, we preserve the original frame order and select the top T

frames with the highest scores for the next stage. The number of selected frames T is constrained to be less than the total number of frames in a gait phase, thereby avoiding redundant information.

2) *Step 2: Training the Gait Classification Model*: Unlike FilterNet, which evaluates frames independently, the gait classification module learns spatiotemporal patterns from sequences of frames. Although both models are trained on the same dataset, the classification module receives only the diagnostically relevant key frames selected by FilterNet, rather than the full video. This selective strategy enhances classification performance by focusing on high-saliency motion features while reducing noise and redundancy.

The selected key frames are passed into the **gait classification** model (see Figure 2 E), which integrates phase-aware motion patterns to identify gait impairments. By capturing essential temporal dynamics, this two-step pipeline improves diagnostic accuracy and supports more interpretable clinical decision-making.

IV. EXPERIMENT

TABLE I: Dataset details.

Label	Diseases detail name	Patient	Video	Total
ASD	Adult Spinal Deformity	54	1046	1046
DHS	Dropped Head Syndrome	16	587	587
LCS_HipOA	Lumbar Canal Stenosis	9	260	324
	Hip Osteoarthritis	2	64	

A. Gait Video Dataset

To support gait and spinal posture analysis, we constructed a clinical video dataset based on previous studies [2], [3], [9], [10]. The study was approved by the University of Tsukuba Hospital Ethics Committee (H30-087) and conducted under the Declaration of Helsinki. The dataset includes 1,957 clips (2–10 seconds each) from 81 patients (61 males, 20 females; age 22–84, mean 70), recorded at 30 fps in 1920×1080 resolution. Videos capture natural walking using fixed-angle cameras. Gait labels were assigned by spine surgeons based on X-ray and clinical evaluation.

We focused on four conditions: adult spinal deformity (ASD), lumbar canal stenosis (LCS), dropped head syndrome (DHS), and hip osteoarthritis (HipOA). LCS and HipOA were merged into a single class (LCS_HipOA) due to similar gait characteristics. To address class imbalance, we adopted an oversampling strategy [11], and ensured subject-wise train/validation split to prevent data leakage.

B. Implementation Details

1) *FilterNet*: To validate the proposed frame selection strategy, we trained **FilterNet** as a standalone classifier across different gait phases. The model adopts ResNet-101 [12] as the backbone, initialized with ImageNet weights and fine-tuned on our gait dataset. We independently trained on four subsets: left swing, right swing, whole-cycle, and mixed-phase (left + right swing) frames.

TABLE II: Evaluation metrics (%) for our baseline experiments. We undertake a comparison between the proposed method and the baseline methods.

Models	Accuracy	Precision	F1 Score
Without FilterNet			
CNN [12]	52.56	81.11	54.01
STGCN [13]	60.42	60.22	59.80
CNN LSTM [14]	38.83	45.41	34.90
Two stream [15]	54.89	76.49	56.46
3D CNN [2]	66.30	73.26	68.55
PhaseMix + 3D CNN [3]	71.43	72.80	71.15
With FilterNet			
FilterNet → CNN LSTM [14]	41.61	40.78	39.94
FilterNet → Two stream [15]	58.09	80.49	60.70
FilterNet → 3D CNN [2]	70.21	71.49	68.43
FilterNet → PhaseMix + 3D CNN [3]	74.52	75.20	74.26

Training was conducted using PyTorch 1.13.1 on a single NVIDIA A5000 GPU (24 GB). Input images were resized to 512×512 , with batch size set to 1 due to memory limits. We used the Adam optimizer with a learning rate of 10^{-5} and early stopping (patience=3) based on validation loss. Evaluation followed the same phase-specific frame selection as training to ensure consistency.

2) *Gait Classification*: To assess overall classification performance, we compared **FilterNet**-enhanced models with several baselines. FilterNet first selects diagnostically relevant frames, which are then input to downstream classifiers (e.g., 3D CNN) to capture inter-phase motion cues.

All methods share identical train/validation protocols. We report accuracy, precision, and F1 score, averaged over 3-fold cross-validation. Confusion matrices are also provided to analyze per-class performance and common misclassification patterns.

V. RESULTS AND DISCUSSION

A. Comparative Study

Table II presents the quantitative comparison between baseline models and those augmented with the proposed **FilterNet**. The results clearly demonstrate that integrating phase-aware frame filtering improves the performance of most backbones, particularly in terms of classification accuracy and F1 score.

Without FilterNet, the best-performing model is *PhaseMix* + 3D CNN, achieving an accuracy of 71.43% and an F1 score of 71.15%. However, when FilterNet is applied prior to this model, performance improves further, reaching **74.52%** accuracy and **74.26%** F1 score — the highest across all settings. This confirms that phase-guided frame selection enhances the discriminative power of the classifier by focusing on diagnostically informative frames.

Interestingly, while the CNN baseline achieves the highest precision (81.11%) without filtering, it suffers from low recall, as indicated by a relatively modest F1 score (54.01%). In contrast, models with FilterNet exhibit a more balanced trade-off between precision and recall, resulting in higher overall F1 scores. For example, *FilterNet* → *Two Stream* achieves 60.70% F1 score compared to 56.46% without filtering.

These findings suggest that **FilterNet** improves both classification robustness and generalization by discarding redundant

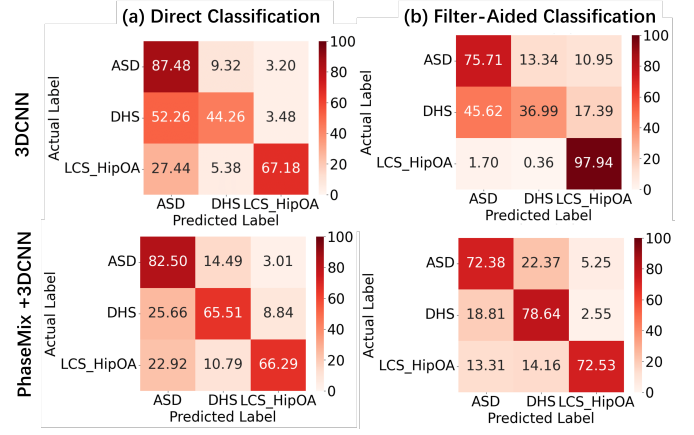


Fig. 4: Confusion matrices comparing (a) direct classification and (b) filter-aided classification. Filter-based frame selection improves class separation, especially for DHS and LCS_HipOA.

or less-informative frames, leading to more compact and diagnostically relevant representations.

Figure 4 presents the confusion matrices for (a) direct classification and (b) classification with the proposed filtering method, using both 3D CNN and PhaseMix-based architectures. These results visualize the classification performance across diagnostic categories and highlight the effectiveness of phase-aware frame selection. We focus on comparing models built upon a 3D CNN backbone, as it has demonstrated superior performance in capturing spatiotemporal gait features. The PhaseMix model further integrates filtered gait phases, enabling more discriminative predictions by leveraging diagnostically salient frames. This comparison illustrates how the proposed FilterNet enhances class separability and improves recognition accuracy, particularly in complex categories.

For the 3D CNN model, the filter-aided strategy leads to substantial improvements in recognizing **LCS_HipOA** (from 67.18% to 97.94%) and **DHS** (from 44.26% to 36.99%, with reduced confusion). This suggests that frame selection based on discriminative gait phases enables the model to better capture subtle class-specific characteristics, especially for underrepresented or motion-ambiguous conditions. In the case of PhaseMix + 3D CNN, the filtered input further enhances classification of **DHS** (from 65.51% to 78.64%) and **LCS_HipOA** (from 66.29% to 72.53%). While **ASD** classification slightly declines in both filtered models, this drop is offset by notable gains in minority class performance, indicating that proposed method promotes more balanced predictions. The results also suggest that direct classification models tend to overfit to dominant classes (e.g., ASD), whereas the phase-aware filtering mitigates this issue by guiding the network’s attention toward diagnostically meaningful motion segments.

In summary, the proposed filter-aided classification framework enhances class separation and improves recognition of complex and underrepresented gait disorders, particularly **DHS** and **LCS_HipOA**, thus supporting its effectiveness in

real-world clinical diagnosis scenarios.

B. Ablation Study 1: Comparative Analysis of Gait Phase Strategies in FilterNet Training

TABLE III: Performance comparison (%) across different training-phase strategies and their impact on downstream classification using PhaseMix 3D CNN aided by the trained FilterNet.

Models	Accuracy	Precision	F1 Score
(A) Step 1: FilterNet Training (Evaluated on Keyframe Prediction)			
Left-Swing Frames	62.26	64.23	61.77
Right-Swing Frames	77.36	81.47	77.18
Whole-Cycle Frames	54.86	57.45	55.26
(B) Step 2: Downstream Classification with PhaseMix + 3D CNN			
Left-Swing Filter	73.23	73.87	73.22
Right-Swing Filter	64.11	67.74	62.94
Whole-Cycle Filter	61.44	75.52	59.84
Mixed-Phase Filter (L+R)	74.52	75.20	74.26

Table III compares various gait-phase-specific training strategies for **FilterNet** and their downstream impact when integrated with PhaseMix and a 3D CNN classifier.

In **Step 1**, filters trained on the **right-swing phase** yield the highest F1 score (77.18%) for keyframe prediction, suggesting that this phase contains discriminative motion cues optimal for relevance scoring. In contrast, training on the **whole-cycle** data leads to the lowest performance (55.26%), likely due to the inclusion of redundant or noisy frames that obscure informative patterns. In **Step 2**, where FilterNet outputs are used to guide downstream classification, the **mixed-phase filter** (combining left and right swing) achieves the best performance across all metrics (F1: 74.26%), highlighting the complementarity of both swing phases in capturing class-specific dynamics. Notably, the left-swing filter also performs competitively (F1: 73.22%), reinforcing that swing phases are more informative than full-cycle sampling. Interestingly, the whole-cycle filter obtains the highest precision (75.52%) but suffers from lower overall accuracy and F1, indicating possible overfitting to fewer confident but less generalizable frames.

Figure 5 further illustrates how each phase-specific filtering strategy affects per-class predictions. The **whole-cycle filter** shows high sensitivity for ASD (90.51%) but underperforms on LCS_HipOA (68.40%), suggesting limited phase specificity. The **mixed-phase filter** balances class-wise performance more effectively, particularly improving predictions for DHS (78.64%) and LCS_HipOA (72.53%). The **left-swing filter** achieves extremely high LCS_HipOA precision (96.41%) but reduces ASD accuracy (60.28%), indicating that LCS_HipOA-specific motion cues are more salient in this phase. Conversely, the **right-swing filter** enhances ASD classification (82.23%) but introduces confusion for LCS_HipOA and DHS.

These results indicate that phase-aware filtering improves not only overall classification accuracy but also inter-class separability. By isolating distinct motion segments, each gait phase contributes unique discriminative cues, motivating future work on adaptive phase fusion for more robust classification.

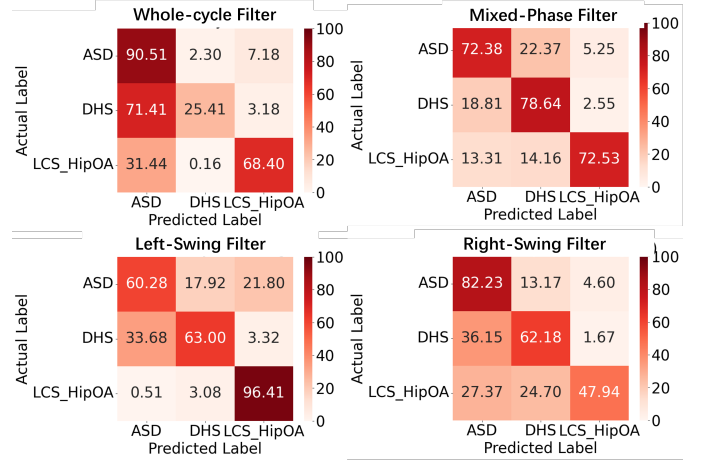


Fig. 5: Confusion matrices of classifiers using **FilterNet** trained on different gait phases: (a) whole-cycle, (b) mixed-phase, (c) left-swing, and (d) right-swing. These results correspond to the classification outcomes reported in Table III (B).

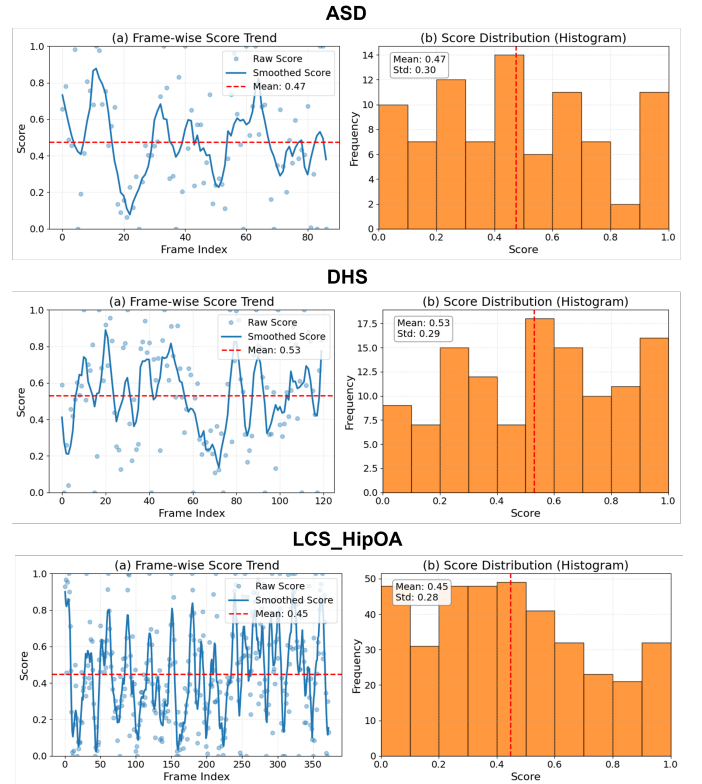


Fig. 6: Visualization of frame-level scores produced by **FilterNet** across three different patient groups. Higher variance or peaked scores in certain groups indicate potential phase-specific discriminative patterns learned by FilterNet.

C. Ablation Study 2: Quality of Filtered Frames

We evaluate the selected frames through visual and quantitative analyses. As shown in Fig. 6, FilterNet assigns frame-wise saliency scores across patient groups. Each row shows (a) the

TABLE IV: Comparison of Frame Selection Methods across Diseases using SSIM, CKPT Motion, and Optical Flow

Disease	SSIM $\uparrow$			CKPT Motion $\uparrow$			Optical Flow $\uparrow$		
	All	Uniform	Ours	All	Uniform	Ours	All	Uniform	Ours
ASD	0.92	0.91	0.91	17.11	19.78	19.92	2.18	1.94	1.83
DHS	0.89	0.87	0.88	26.89	29.25	31.00	4.55	4.68	4.91
LCS_HipOA	0.92	0.88	0.91	11.16	14.37	20.47	3.70	3.17	3.50

temporal score trend and (b) its distribution, revealing group-specific patterns that highlight phase-dependent discriminative cues.

Quantitatively, models trained using top-k filtered frames consistently outperform those using uniformly sampled or low-score frames in terms of classification accuracy and F1 score. Qualitative inspection further supports this, revealing that high-score frames frequently align with dynamic motion phases, such as mid-swing or compensatory trunk movements, whereas low-score frames often correspond to static or redundant postures. These findings suggest that our learned filter not only enhances model performance but also offers interpretable cues about clinically informative gait phases.

To capture temporal motion variations, we evaluate frame-wise movement intensity using three complementary metrics: (1) Structural Similarity Index (SSIM) to assess spatial consistency, (2) Keypoint-based motion (CKPT) to measure postural change, and (3) Optical flow magnitude to reflect pixel-level displacement between adjacent frames. For both metrics, we compute the average across three frame selection strategies: all frames, uniformly sampled frames, and our proposed filtered frames. This combination provides a simple yet effective approximation of motion strength at the frame level, without requiring dense annotations or labels.

As summarized in Table IV, our filtered frame selection consistently exhibits higher CKPT motion than the all-frame baseline, while maintaining comparable SSIM values, indicating that it captures more pronounced postural variations without sacrificing spatial coherence. Although all-frame sampling tends to yield the highest optical flow—largely due to temporal redundancy—it may include visually repetitive or clinically less informative frames. In contrast, our method achieves a better trade-off between motion richness and visual compactness, resulting in a more concise and discriminative subset of frames for downstream analysis.

VI. CONCLUSIONS

In this study, a phase-aware frame filtering framework, FilterNet, was presented to enhance spatiotemporal representation learning for periodic motion analysis. By selecting phase-relevant frames, the method reduces temporal redundancy and emphasizes meaningful motion variations. Experimental results on a clinical gait dataset demonstrate consistent improvements in classification accuracy and interpretability through the identification of key motion transitions. Although validated in a clinical gait analysis setting, the proposed framework can be extended to other structured motion understanding tasks. Future research will focus on integrating FilterNet with

transformer-based temporal models and adapting it to in-the-wild conditions.

ACKNOWLEDGMENT

This work was supported by MEXT Promotion of Development of a Joint Usage/Research System Project: Coalition of Universities for Research Excellence Program (CURE) Grant Number JPMXP1323015474.

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